

■ AI-Driven Drug Response Prediction & Knowledge Graph

End-to-end scientific AI project demonstrating survival analysis, Bayesian inference, NLP, and biomedical knowledge graphs with an interactive dashboard.

■ Workflow

Synthetic Data → Survival Analysis (Kaplan–Meier, CoxPH) → Bayesian Modeling (PyMC) → Knowledge Graph (NetworkX) → NLP (Transformers + LangChain) → Streamlit Dashboard

■ Highlights

- Survival Analysis: Kaplan–Meier & CoxPH models on drug response data
- Bayesian Modeling: PyMC credible intervals for treatment effects
- Knowledge Graph: drug–target–disease graph (Neo4j-ready, GNN-ready)
- NLP Stub: HuggingFace + LangChain for biomedical literature Q&A;
- Interactive Dashboard: Streamlit app for clinicians & stakeholders
- CI/CD: GitHub Actions with automated pytest

■ GitHub Repo:

<https://github.com/Rutvijmw/Ai-life-sciences-project>

